ELEC40006 Electronics Design Project

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1 Abstract

The EEELunarRover is a remotely controlled rover designed to survey an artificial lunar surface and classify rocks by age and type. Rock age is decoded from an 89 kHz ASK radio signal, while rock type is identified using infrared pulse rate, 40 kHz ultrasound detection, and magnetic polarity. The rover also had to remain highly manoeuvrable while satisfying the mass constraint of less than 750 g.

Type	**Infrared**	**Ultrasound**	**Magnetic**
Basaltoid	$\lambda = 547\ \mathrm{s}^{-1}$	40 kHz	Down
Gravion	$\lambda = 312\ \mathrm{s}^{-1}$	–	Down
Regolix	$\lambda = 312\ \mathrm{s}^{-1}$	40 kHz	Up
Lunarite	$\lambda = 547\ \mathrm{s}^{-1}$	–	Up

Table 1: Rock classification matrix based on infrared, ultrasound, and magnetic readings.

2 Project Management

2.1 Team Responsibilities

At the start of the project, the rover was decomposed into three interacting development areas: sensing, hardware, and software. This structure was chosen because the main design challenge was not only to produce working individual circuits, but to integrate sensors, mechanical packaging, motor control, and the user interface into a single rover within the mass limit of 750 g. The division also made subsystem interfaces explicit, which reduced the risk of late-stage integration failures.

- **Sensing:** This area covered the rock-age and rock-type detection subsystems. The age-detection system required an 89 kHz ASK radio receiver, while type classification required infrared pulse-rate measurement, 40 kHz ultrasound detection, and magnetic-polarity detection. These sensors had to be electrically reliable, physically mountable on the chassis, and simple enough to integrate into the final classification algorithm.
- **Hardware:** This area covered the mechanical and electrical construction of the rover. A custom mecanum-wheel chassis was selected to maximise manoeuvrability on the artificial lunar surface while leaving sufficient mounting space for sensors, PCB/stripboard circuits, wiring, and batteries. The chassis was 3D-modelled and printed, with layout decisions constrained by sensor viewing angles, circuit dimensions, wiring access, centre of mass, and the 750 g mass limit.
- **Software:** This area covered the rover control, communication, and user-interface systems. The onboard software processed sensor inputs, generated motor-control outputs, and communicated with the external interface. The UI displayed telemetry and rock classifications while allowing real-time remote control of the rover.

Responsibilities were assigned according to each member's technical strengths, but integration-critical areas were deliberately shared. This was important because several design decisions affected multiple subsystems; for example, PCB dimensions affected chassis design, sensor placement affected both classification reliability and mechanical layout, and motor-control choices affected power distribution and software control.

Member	**Role(s)**	**Responsibilities**
Fred Ayres	Electrical engineer	Magnetometer, IR sensing design, implementation and testing, stripboard design, construction and testing, power system
Owais Darwesh	Electrical engineer, editor	Radio receiver, report writing
Sijie Fei	Software engineer	Communication, control system, user interface
Joey Liu	Electrical engineer	Ultrasound sensing, PCB design, construction and testing
Colin Qian	Mechanical engineer, electrical engineer, editor, team leader	3D design and assembly, PCB design, IR sensing design, implementation and testing, power system, communication, control system, report writing, project management
Ximan Wang	Electrical engineer, editor	IR sensing design, presentation preparation
All members	Integration and testing	Subsystem integration, debugging, final testing

Table 2: Team members, roles, and responsibilities.

2.2 Management Methods

The project was managed using weekly structured meetings, short-term task allocation, shared documentation, and explicit tracking of subsystem dependencies. This approach was selected because the rover contained several parallel workstreams whose outputs had to converge before final integration. In particular, the mechanical design depended on sensor dimensions and PCB/stripboard layout, while PCB ordering depended on the sensor circuits being tested first.

After each meeting, short-term objectives were assigned to individual members according to the task allocation in Table 2. Objectives were usually set either for the following week or for a specific milestone, such as the interim presentation or PCB-order deadline. This made the project plan responsive: tasks could be reprioritised when faults were found, rather than continuing with an invalid design.

Project constraint	**Management implication**	**Evidence / outcome**
750 g mass limit	Heavy components and chassis choices had to be tracked before final assembly.	A shared spreadsheet was used to monitor mass; the running estimate remained below the limit, with the spreadsheet showing approximately 489.9 g during planning.
Budget limit	Component choices had to be checked before ordering, and spare components had to be prioritised.	The spreadsheet tracked spending, with approximately £47.82 recorded during planning. Remaining budget was used to buy spare op-amps and comparators, which was useful after an op-amp was damaged during soldering.
Sensor-dependent chassis design	Mechanical layout could not be finalised until sensor positions, PCB size, and wiring routes were known.	The initial chassis was completed on 13 May, but finalisation was delayed until 5 June to incorporate updated circuit and sensor constraints.
PCB lead time	PCB ordering required sufficient confidence that the sensing circuits were functional.	The PCB order was paused after the IR circuit failed during testing, avoiding the cost and delay of manufacturing an incorrect board.
Final integration deadline	Backup routes were needed for components with delivery or reliability risks.	Stripboard space, a motor-driver fallback, and sensor-classification fallback logic were prepared to reduce dependence on any single subsystem.

Table 3: Relationship between top-level project constraints and management decisions.

The following tools were used to support project management:

- **Notion:** Notion was used as the central workspace for task tracking, documentation, deliverables, and progress monitoring. A Gantt chart was used to show deadlines, subsystem dependencies, and integration milestones. Each Gantt entry contained supporting records, files, and progress notes, which made the project history traceable during report writing and helped the team identify which tasks were blocking others. This was particularly useful for PCB design, where the layout depended on the completion of the infrared, ultrasound, magnetic, and radio-receiver circuits.
- **WhatsApp:** WhatsApp was used for rapid day-to-day communication, including design updates, short-term reminders, and debugging discussions. It was effective for urgent

communication, but less reliable as a task-management system because important reminders could be lost in the message history. A weakness of the management process was therefore that some short-term actions were communicated informally rather than always being converted into Notion tasks. In future work, every action agreed on WhatsApp should be added to the shared task list with an owner and deadline.

- **Budget and weight spreadsheet:** A shared spreadsheet was used to track component cost and estimated mass. This allowed design choices to be checked quantitatively against the budget and weight constraints before ordering or assembly. It also helped justify the decision to buy spare op-amps and comparators using the remaining budget, since this reduced the risk of a single soldering or testing failure blocking progress.
- **KiCad:** KiCad was used to design and review schematics and PCB layouts before construction or ordering. This ensured that the electrical design was documented and could be checked by multiple team members. However, version control was not fully centralised: some updated schematic files were shared through WhatsApp, creating the risk of version conflicts. In future projects, all KiCad files should be stored in Github, with clear version control and forks rather than outdated versions circulated informally.

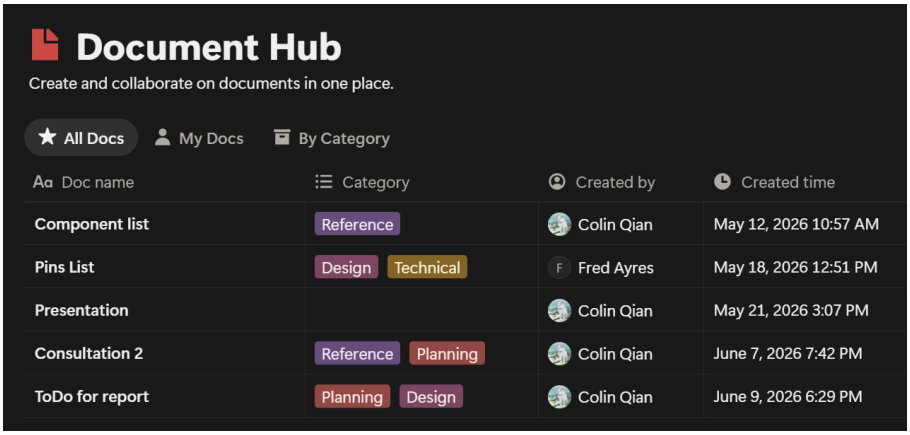


Figure 1: Shared document hub in Notion.

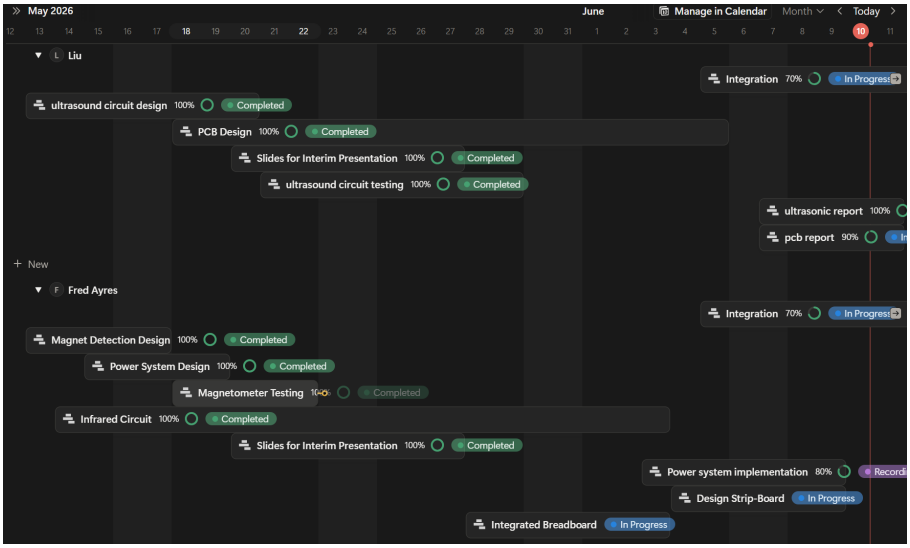


Figure 2: Notion Gantt chart used to track task progress, deadlines, and subsystem dependencies.

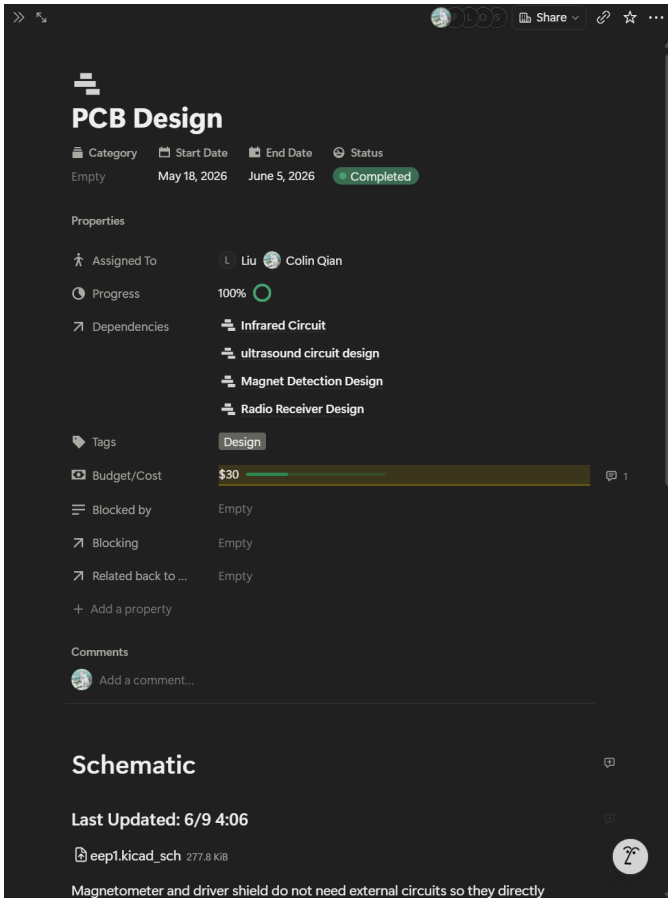


Figure 3: Example Gantt entry for PCB design, showing dependencies on the sensing subsystems.

Total spent	47.82		Planned spending	0.00							
Total weight	489.9										
Group	Name	price	weight/g	total weight	Count	Source	Status	Order from	Order code	Spent	Note
Hardware	Yellow motor	0.00	30	119.8	4	eeebug	Using				
	Mecanum wheel	0.00	50	197.2	4	3d print	Using				60mm, pla
	Rivet	3.95	-	-	50	Rapid(shop)	Using	Rapid	86-1564	3.95	shaft
	Chassis	0.00	-	-	1	3d print/laser cut	Using				pla
	adafruit & shield	0.00	94.3	39.3	1	extension kit	Using				
	battery	0.00	26.1	104.4	4	eeebug	Using				
	battery holder	0.00	14.6	14.6	1	eeebug/3d print	Using				
	pcb	5.00	-	-	1	shop	Using			0	independent budget
	Sensors	0.00	14.6	14.6	1	Circuit	Using				
Circuit	SL0190 -- CA3140	2.72			1	shop	Purchased	eedstores	SL0190	2.72	opamp
	SI0160 -- LM311N	0.54			1	shop	Purchased	eedstores	SI0160	0.54	opamp
	MCUSD16A40S12RO	4.31			1	external	Using	onecall	SN36976	1.78	ultrasonic transducer
	SEN4019(*SEN0419)	6.07			1	Farnell(shop)	Using	onecall	4308272	5.06	3 axis magnetometer
	WPSH207	11.63			1	Farnell(shop)	Discard	Conrad	2481967 - 62	11.63	Motor Driver Shield
	MCP6022 I/P x5	8.40			1	RS components	Using	RS	403-064	8.4	opamp
	SFH 314 FA	3.43			10	Onecall(shop)	Using	onecall	2981735	3.43	infrared sensor
	MCP6022 I/P x3	1.47			3	Onecall(shop)	Using	onecall	9758640	4.41	opamp
	Stripboard	9.11			1	RS components	Using	RS	100-4328	9.11	stripboard
	MCP6022 I/P x3	1.47			3	Onecall(shop)	Purchased	onecall	9758640	4.41	opamp
	MCP6541	4.01			1	RS components	Purchased	RS	402-790	4.01	comparator

Figure 4: Budget and weight spreadsheet used to monitor cost and mass during design.

2.3 Delays and Risk Management

Risk management was necessary because many subsystem tasks were sequentially dependent. The main risks were late component delivery, untested circuits delaying PCB ordering, and mechanical redesign caused by changing sensor or circuit dimensions.

The initial chassis design was completed on 13 May. However, construction was delayed because final sensor positions, circuit dimensions, and PCB status were not yet confirmed. Since the chassis had to provide accurate mounting points for the sensors and sufficient internal

space for electronics and wiring, manufacturing the first design immediately would have risked producing a chassis that was incompatible with the final electronics. The chassis was therefore revised as the sensing and PCB designs developed, and was finalised on 5 June. Although this reduced mechanical build time, it improved the probability that the final chassis would support successful subsystem integration.

The PCB schematic was first produced on 25 May, but ordering was paused after the IR circuit failed during testing. This was a deliberate management decision: ordering the PCB before resolving the IR fault would have risked manufacturing a board containing an incorrect or unreliable sensing circuit. The schematic was revised and finalised on 6 June, the PCB layout was completed on 7 June, and the board was ordered on 8 June. This delayed the PCB schedule, but reduced the risk of a more serious late-stage failure.

The motor shield was ordered on 21 May, but delivery was repeatedly delayed. To avoid dependence on a single component, the team prepared an alternative motor-control route using Arduino outputs and dual H-bridge motor-driver circuitry. This contingency became essential because the motor shield did not arrive in time. The fallback motor-driver circuit allowed drive-system development to continue and prevented the delayed delivery from stopping the rover from operating.

A weakness in the original project plan was that module-level testing milestones were not specified in enough detail at the start. As a result, faults such as the IR-circuit failure were discovered later than ideal, compressing the available time for integration and debugging. A better plan would have included formal test gates for each sensing subsystem before PCB layout, followed by a separate full-system integration gate. This would have made circuit readiness more objective and reduced the risk of late design changes.

Risk	**Impact**	**Mitigation**	**Evaluation**
IR circuit failed during testing	PCB order could have been based on an incorrect sensing circuit.	PCB ordering was paused until the IR circuit was revised and retested.	This delayed PCB production, but avoided committing to a faulty board design.
Motor shield delivery delay	Rover drive system could not rely on the planned shield.	A dual H-bridge motor-driver fallback was prepared.	The fallback was necessary because the shield did not arrive in time, allowing the rover to remain operational.
Changing sensor and PCB dimensions	Chassis mounting points and internal layout could become incompatible with final electronics.	Chassis finalisation was delayed until sensor and circuit constraints were clearer.	This reduced available build time, but improved mechanical-electrical compatibility.
Faulty or damaged components	A single damaged op-amp or comparator could block testing.	Spare op-amps and comparators were purchased using remaining budget.	This was useful after an op-amp was damaged during soldering.
Loss of working sensor input	Rock-type classification could fail if one sensing module was unreliable.	A software classification fallback was considered using the remaining two sensors where possible.	This reduced the probability that one sensor fault would make all classification impossible, although it could reduce confidence for ambiguous cases.

Table 4: Risk management decisions and their effect on the final project.

2.4 Backup Plans

To reduce the impact of component failures or delivery delays, several backup options were prepared. These alternatives allowed the team to continue development even if the preferred implementation became unavailable.

Subsystem	**Backup option**	**Implementation method**
PCB	Stripboard circuit	123.2 cm^2 reserved on the chassis
Motor shield	Motor-driver circuit	Separate stripboard-based driver circuit using dual H-bridge motor drivers
Rover body	EEEBug platform	Existing wheels and chassis used as a fallback platform
IR, ultrasound, or magnetic sensor	Software classification fallback	Infer rock type using the remaining two working sensors where possible

Table 5: Backup plans for key rover subsystems.

These backup plans were particularly important because the project contained several interdependent subsystems. For example, a delay in PCB production could affect mechanical integration, while failure of one sensing module could affect the rock-classification algorithm. By reserving chassis space for stripboard circuits and preparing alternative motor-control methods, the team reduced the risk of a single component failure preventing the rover from operating during the final demonstration.

The most significant example was the motor shield. Because the shield did not arrive due to continued factory delivery delays, the motor-driver fallback was not only a theoretical backup but became part of the final implementation route. This demonstrates that the risk-management process had a direct effect on the success of the final rover, rather than being a purely administrative exercise.

Overall, the management process was effective because it linked planning decisions to technical constraints: mass, budget, PCB readiness, sensor integration, and final demonstration reliability. The main limitation was that testing gates were introduced informally rather than being fully defined from the start. For future projects, the strongest improvement would be to add explicit subsystem test milestones before PCB ordering and before mechanical finalisation, with each test producing quantitative pass/fail evidence. This would make the transition from design to implementation more controlled and reduce the risk of late integration faults.